The empirical recurrence for OEIS A283662, recovered from its tail

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Abstract

OEIS A283662 records "Empirical recurrence of order 84" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 752$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 11$, so the recurrence is proved for every $n \geq 95$. Its last coefficient is $c_{84} = 27 \neq 0$, so no further tail satisfies anything shorter; any order-84 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A283662 is "Number of nX4 0..1 arrays with no element unequal to more than four of its king-move neighbors, with the exception of exactly two elements, and with new values introduced in order 0 sequentially upwards.". It has offset 1 and begins

$$0, 0, 403, 11886, 165152, 1712052, 15351085, 126810474, \ \ldots$$

The entry states:

> Empirical recurrence of order 84 (see link above)

As of the "Last modified" line on the live entry (Mar 13 2017 09:52:55, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \ = \ \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 752$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 752$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \ldots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a. $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 84: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 11$ is the first whose minimal order is exactly the stated 84.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 1504$ exact terms from $n = 11$ on, it returns order 84 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{84} c_i\, a(n-i),$$

c_1	15	c_{22}	538967295	c_{43}	517955298	c_{64}	409080
c_2	−33	c_{23}	−1159917015	c_{44}	−315164580	c_{65}	567324
c_3	−430	c_{24}	327856021	c_{45}	−366899900	c_{66}	939559
c_4	1551	c_{25}	1997292246	c_{46}	−23616435	c_{67}	828483
c_5	6492	c_{26}	−2395998588	c_{47}	275128845	c_{68}	542766
c_6	−26665	c_{27}	−1105135703	c_{48}	138611054	c_{69}	48057
c_7	−59847	c_{28}	4023572214	c_{49}	−130502376	c_{70}	−82968
c_8	258870	c_{29}	−2050434309	c_{50}	−67712820	c_{71}	−132624
c_9	331881	c_{30}	−2946610073	c_{51}	60838779	c_{72}	−86068
c_{10}	−1598496	c_{31}	5503690851	c_{52}	−11046309	c_{73}	−50904
c_{11}	−771354	c_{32}	−654141900	c_{53}	−60984414	c_{74}	−8703
c_{12}	6292581	c_{33}	−6802263955	c_{54}	−13113096	c_{75}	2052
c_{13}	−2767071	c_{34}	3963584502	c_{55}	23586834	c_{76}	8757
c_{14}	−13429335	c_{35}	5696657778	c_{56}	19184235	c_{77}	5481
c_{15}	28937198	c_{36}	−4822815492	c_{57}	10087278	c_{78}	1458
c_{16}	−5090430	c_{37}	−3653728581	c_{58}	7091943	c_{79}	−243
c_{17}	−98970153	c_{38}	3645091911	c_{59}	1448109	c_{80}	54
c_{18}	135233406	c_{39}	1949058418	c_{60}	−2382577	c_{81}	−243
c_{19}	125688249	c_{40}	−2044845855	c_{61}	−5474568	c_{82}	−162
c_{20}	−424655082	c_{41}	−966689568	c_{62}	−5004564	c_{83}	0
c_{21}	215104468	c_{42}	939759620	c_{63}	−2504206	c_{84}	27

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 95$, and it is the recurrence OEIS A283662 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 11$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{84} - c_1 x^{84-1} - \cdots - c_{84}$. Its constant term is $-c_{84} = -27$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 84 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 84, so $\deg q = 84$, and it is monic. It holds from some index t on. From $\max(t, 11)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 84, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 19 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 84, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 27.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-84 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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